The Starter Guide for dbt™

A free guide to help you optimize your new dbt project.

If there is one tool I *always recommend* learning (and implementing) in the modern data stack, it's dbt (data-build-tool).

Not only is it powerful as a **data transformation tool** but is also supported by an amazing **open-source community** that has helped 10s of thousands of data engineers (including me) better approach modern data platforms.

But dbt is also relatively new and can be a bit confusing, especially when first getting started.

In this guide, you'll learn:

- What exactly is dbt and why is it so popular?
- 5 best practices to consider
- A 10-part checklist to help you start your first project

Let's dive in.

WHAT IS DBT?

dbt is a code-based data transformation tool designed for the modern age of data.

In traditional data pipelines, the data **transformation** step is usually handled with drag/drop tools and an "ETL" pipeline optimized for row-based databases.

However, the advances in cloud computing and the growing expectations of data teams has forced us all to rethink the way we approach this entire process.

Version control & automation is a must, data quality is more important than ever and data is now treated like a standalone product rather than an after thought.

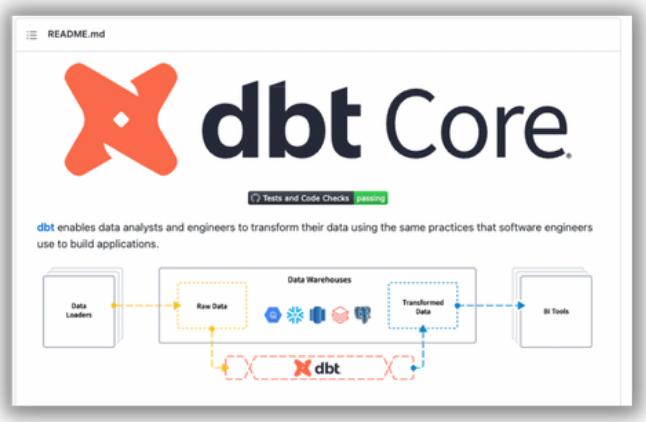
Enter dbt.

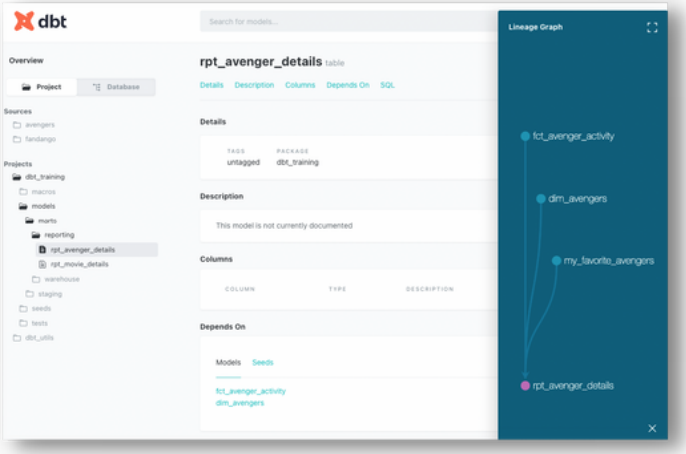
dbt allows you to create data transformation logic using SQL but in a way that can be incredibly dynamic.

This is accomplished by using features like **Jinja templating** and **yaml-based config files**.

At a high level, dbt uses these features to compile SQL queries which are then sent to your cloud database server to be executed. All of this happens within a single "run".

The result is new, transformed data models (tables, views) that are ready for analytics.





You can also add tests, create a documentation website and automate the entire process with just a few simple commands.

This is just scratching the surface and there are many subtleties that are outside the scope of this starter guide.

But what about all of this has made dbt skyrocket in popularity?

WHY IS IT SO POPULAR?

Functionality plus an opinionated approach solve data workflow problems.

We know that data now needs to be managed like a standalone product.

But even if that makes sense at a high level, it's hard to just start changing things without the right processes and tools in place. Not to mention this is a new thought process for most data folks and there are many questions around best practices.

dbt is so popular because it is not only clear on their viewpoint but provides the tool and community support to help you along the way.

It also currently sits alone as the clear best-option for this component of the modern stack.

With the right tools and the right approach, we can better solve this workflow problem so that your data team starts looking like a software engineering team full of automation, documentation and consistency.

5 BEST PRACTICES TO CONSIDER

A breakdown of common scenarios and approaches for optimizing dbt.

TIP #1: USE CTES

CTEs (Common Table Expressions) are treated as passthroughs

At the end of the day, all dbt "models" are ultimately just SQL statements. And traditionally, one would think that a **select** * from anything is a bad idea - but not so fast.

Modern cloud databases are really smart and treat CTEs as pass-throughs. They only care about the columns you select at the very end and will disregard all others completely.

For that reason, it is suggested to use CTE's as "imports" and to overall organize your code.

For example, a typical dbt model might look something like this:

For more on this topic, check out [this article](#).

models > marts > warehouse > fct_movie_voting.sql

```sql
with fandango_scrape as (

    select * from {{ ref('stg_fandango__scrape') }}

),

fandango_scores as (

    select * from {{ ref('stg_fandango__score_comparison') }}

),

final as (

    select
        fandango_scores.movie_id,
        fandango_scrape.scrape_vote_count,
        fandango_scores.metacritic_user_vote_count,
        fandango_scores.imdb_user_vote_count,
        fandango_scores.fandango_votes,
        fandango_scores.fandango_difference

    from fandango_scores

    left join fandango_scrape
        on fandango_scores.movie_id = fandango_scrape.movie_id

)

select * from final
```

TIP #2: CREATE A STYLE GUIDE

Consistency will make or break a project

Anytime you get into a group, you're bound to have different approaches. This is not necessarily a bad thing, but it makes consistency hard to achieve.

The best way to combat this is to get everyone on the same page from the start. A great way to do this is to create a style guide for your project.

This should include things like how you plan to name your models *(e.g. prefixes, plural vs singular)*, how you will name columns *(e.g. snake vs camel case, is_ for boolean)* or how you're write your queries *(e.g. all lowercase, line length limits)*.

More specifically, you should create a separate .md file just for your style guide and reference it from your main README.md for others to easily find.

Check out [this example](#) style guide from dbt to get you started.

TIP #3: LEVERAGE DIRECTORIES

Remember DRY (dont-repeat-yourself)

Dbt is designed for modularity and reusability. One of the best ways you can take advantage of this is through directories.

First, it will it help you organize your project. But more importantly, it will allow you to set directory-level model configurations in your dbt_project.yml file rather than repeating the same configuration in each individual model. Just imagine if you had 100s of them.

```
models:
  dbt_training:
    staging:
      +materialized: view
      +schema: staging
    marts:
      +materialized: table
      warehouse:
        +schema: warehouse
      reporting:
        +schema: reporting
```

The further down in directory layers you go, the more it will take precedence until you finally get to the model itself.

Set things that are reusable at the highest level, and only add new configs as you move further down. However, avoid creating model-specific configs in your dbt_project.yml if possible to avoid over-cluttering it.

For more on model configurations, check out [this link](#).

TIP #4: AUTOMATE TESTING

We're developers, not full-time testers

However...testing is a critical part in creating trust in the final data product. Fortunately, dbt is a command line tool and can be easily automated.

Once you've created your tests *(singular/generic)* in your project, you can use automations like GitHub Actions or GitLab Pipelines to automatically test your code before it gets too far.

For example, every time a new pull/merge request is opened, you can set a workflow to:

1. Deploy your dbt models (dbt run)
2. Test your data (dbt test)

Any failures would block the merge. While nothing is 100% bullet proof, this will allow you to catch obvious errors right away in an automated fashion. The more robust your test suite gets, the better this process will be.

Learn more about [GitHub Actions](#) and [GitLab Pipelines](#)

TIP #5: DON'T HARDCODE TABLES

This works against core dbt functionality

It might be second nature to write out your table names in a SQL query, but you'll want to avoid this at all costs with dbt.

Technically it will still work, but this approach will cause you to miss many of the best features of dbt. Instead, you should always refer to a table in your models by using either the "source" or "ref" Jinja functions.

```
select * from {{ source('avengers', 'avengers_history') }}
```

```
select * from {{ ref('stg_avengers__avengers_history') }}
```

Why? First, using these functions will allow dbt to create dependencies between your sources and models which in turn allows it to create a proper DAG. This means you can easily run any upstream/downstream models using simple operators.

Second, it will automatically set the correct database/schema when running your project in different environments. If it's hardcoded, you'll be stuck with the same one and your code will be much less dynamic.

BONUS: HOST DOCUMENTATION

Give your stakeholders what they want

Stakeholders love data dictionaries, and dbt allows you to easily create a pre-built website through two simple commands:

1. dbt docs generate
2. dbt docs serve

However, you're going to want to find a convenient way for people to access it. You could always keep it on a network server and give people access. But another option is to push it to a hosted provider.

For example, use your cloud provider or a static web hosting service like Netlify or Github Pages.

You can even add a step in your automated workflow (tip 4) to publish the latest version of your docs to one of these locations.

Once that's set up, you can set it and forget it and direct 99% of stakeholder questions to the site.

10-PART CHECKLIST

Let's make sure you set yourself up for success

Once you've initialized a brand new dbt project, it is a feeling of both excitement and intimidation.

The freedom of open source is every developer's dream but unless you've worked with dbt before there's a good chance you'll overlook something subtle.

Use this checklist to help you feel more comfortable that you've set the proper foundations for your project and can get off to a great start!

The List

1. **Create a style guide**
    a. Add it directly in your project or link to it from your readme.md file
2. **Determine high-level model/ directory structure**
    a. Here is a link to a helpful breakdown of a common way to structure it
    b. You'll typically want to match it closely with your database structure
3. **Create a request template on your hosted git provider for pull/merge requests**
    a. This will facilitate your review process and help you catch errors ahead of time
    b. Here is a video where I show you how to do this for GitHub
4. **Set default values as variables in dbt_project.yml**
    a. This avoids the need to hard-code the same values in different places (DRY)
5. **Review packages on dbt hub to avoid reinventing the wheel**
    a. A great place to start is dbt-utils
    b. At the very least, you can use packages for inspiration to create your own macros
6. **Implement hooks for repetitive tasks**
    a. Common examples are for auditing, inserts/updates, permissions, etc.
    b. You can also create and use macros as hooks instead of putting SQL directly
7. **Decide on commit message & branch naming conventions**
    a. Helpful for auditing
    b. An example template is [task-number]-devInitials-description
        i. [JIRA-1234]-mk-create-a-staging-model
8. **Update & make use of model config section in dbt_project.yml**
    a. Assign any reusable configs in this section
9. **Decide if you'll do one yaml file (schema) per directory or one per model**
    a. This will depend on the scope of your documentation and testing
    b. If unsure, start with one per directory and move to individual if it gets too long
10. **Join the dbt Slack Community (it's free!)**

Thanks For Reading!

I hope you found this guide helpful. There is **so much more** to learn about dbt, but hopefully now you have a better appreciation for its purpose and the type of things to expect as a developer.

For more dbt-specific content, check out the rest of my YouTube playlist.

Cheers,
Mike